

NAME: DIVYAANSH VATS

BLUE MACHINES (FORWARD DEPLOYED ENGINEER)

KESTREL HEALTH INSURANCE ANALYSIS

PART-1 (WHAT I FOUND)

Main Metric

1. Containment rate: 71.46%(W1)-->62.3%(W2)-->51.12%(W3). A 20 percent point drop in 3 weeks.
2. Transfer Rate: 28.53%-->37.67%-->48.87%
3. Abandoned rate: 2.3%-->3.2%-->3.5%. It increased slightly, but it is not the main issue.

First Check (Is it the new version causing the problem?)

1. V14 total calls are 1948, and V15 total calls are 1596.
2. In W2, when both versions were running: v14 containment was 66.98% and v15 containment was 55.29%.
3. So there is a real difference between the two versions.
4. But this is not the full story because v15 continued to fall in W3 to 51.12% even though there was no new deployment.

Second Check (Which Intent is Causing the Problem?)

In W2 I compared v14 and v15 for each intent:

Claim_status: 64.9%-->44.9% (20pp gap, it was the largest by far)

Renewal_premium: 67.9%-->58.7% (9.2pp)

Policy_details: 63.9%-->60% (3.9pp)

Network_hospital: 70.3%-->67% (3.3pp)

Claim_status is clearly the main problem.

The other intents changed much less.

What I ruled out

I also checked other possible causes:

1. ASR confidence: 91.8%-->91.4%-->90.4%. It changed very little, so it is not an ASR problem.
2. Tool Failure Rate: 1.51%-->1.08%-->0.96%. The failure rate actually went down while transfers went up, so normal backend/tool errors do not explain the problem.
3. Language: Both English and Hindi declined in a similar way. So this is not a language-specific problem.

4. Call duration in W2: v14=135.4 sec and v15=135.2 sec. They are almost the same, so calls were not simply becoming longer or more confusing.

The mechanism: One intent became slower

Claim_status response time: 445ms(W1)-->634ms(W2)-->950ms(W3)

The other three intents stayed almost flat at 440-465 ms across all 3 weeks.

Backend tool latency for the claim lookup did not change.

However, claim lookups increasingly returned **no claim data**, which caused the agent to retry/rephrase the lookup.

So the problem is **not that the database/API became slower**. The problem is that the **claim lookup increasingly failed to return a usable claim, causing additional agent-side retries and interactions**, which increased the overall response time.

PART-2 (THE REAL TIMELINE)

This is the most important part of the investigation.

August 25: The degradation starts. Daily containment and response time both start getting worse on this date. No BLUE MACHINE deployment happened on August 25.

August 28: Blue Machine released the v14→v15 prompt change. It added claim-status rephrase guidance. Response time became even worse after this change. But the problem had already started 3 days earlier.

This migration changed the format of claim reference numbers for the newly moved policies. This caused the claim lookup to stop matching numbers for the newly moved policies.

OTHER TICKETS SUPPORT THE SAME PROBLEM

AUG 25: Callers had to repeat their claim numbers

AUG 28: A caller had the correct claim number, but the bot said it could not find the claim.

SEP 5: The same problem was still happening, and a human agent found the claim manually.

CONCLUSION:

The evidence points to Kestrel's TPA migration causing the claim reference matching problem. The Blue Machines' prompt change on August 28 made the problem worse, but it did not start the problem.

PART-3 (WHAT IT'S COSTING)

Human agent cost

1. ₹13,986 → ₹18,396 → ₹22,764 per week
2. This is a **63% increase in 3 weeks**.

Customer satisfaction

Transferred calls:

- 4.3 → 4.1 → 3.6

This is a real drop and is a **churn-risk signal**, not just a cost issue.

Contained calls:

- 4.5 → 4.5 → 4.4

This stayed almost stable.

This tells us that **successful calls are not getting worse**. The problem is mainly in the **failure path**.

Operational impact

There are also problems beyond money:

- **August 27:** High-priority ticket says agent queue volume increased sharply.
- **September 4:** A high-priority ticket says the handle time increased and the **wait-time SLA was missed**.

So this is also a **service-level problem**, not only a cost problem.

Kestrel was already concerned.

Kestrel's **Head of CS escalated the issue internally by August 31 (KH-4566)**.

So they were already concerned before this review.

PART-4 (WHAT I DID)

1. I split the data.

I looked at containment and transfers:

- By week
- By agent version
- By intent

I did not only look at the overall number.

2. I checked the containment formula.

I compared Kestrel's reported containment formula with my own calculation from the raw call data.

This confirmed that abandoned calls are not included in the containment denominator.

3. I tested possible causes.

I checked:

- ASR quality
- Tool/backend failure rate
- Language
- Call duration

I used the data to test each one instead of assuming the cause.

4. I found the main mechanism.

I compared response time by intent, instead of only comparing versions.

This showed that `claim_status` was the main problem.

5. I looked at daily data.

The weekly data showed that performance was getting worse, but it did not show the exact start date.

So I checked the data day by day.

This showed that the real problem started on August 25.

That was important because my first guess was the August 28 Blue Machines release.

The daily data showed that my first guess was wrong.

6. I checked the support tickets.

I compared `support_tickets.csv` with the timeline.

This helped me find a Kestrel-side cause that was not present in the Blue Machines deployment log.

This is important:

The evidence points to the TPA migration and resulting claim-reference format change as the initiating cause.

So if I had only looked at `deployments.csv`, I could have wrongly blamed the Blue Machines prompt change.

7. I corrected my first hypothesis.

My first thought was that our August 28 release caused the problem.

But the daily data did not support that.

So I changed my conclusion before presenting the analysis.

I think this is a strength because I followed the data instead of trying to prove my first idea.

PART-5 (WHAT WOULD I TELL KESTREL)

Immediate Today

1. Roll back the August 28 prompt change.

Roll it back now.

It is low risk, removes one active cause of the delay, and gives us some time.

2. Communicate with Kestrel now.

Do not wait for the review meeting.

Tell them:

- We found the cause.
- It is linked to their TPA migration.
- Here is what we will do.
- Here is when we will do it.

3. Set the right expectation.

Be clear that the rollback will reduce the problem but will not fully fix it.

This avoids a situation where Kestrel expects the problem to disappear completely after the rollback.

This week

4. Work with Kestrel IT

We need to work with Kestrel IT, not only Ops or CS.

We need to update the claim lookup so it can handle the new TPA claim-reference format.

For this, we need Kestrel's format specification.

Blue Machines cannot complete this part alone.

5. Add a quick fallback.

If a claim lookup fails because of the new format:

Transfer the call immediately instead of making the agent try again or rephrase the request.

A fast and clear failure is better than a slow and confusing conversation.

Long-term fix

6. Notify Blue Machines before backend changes.

Kestrel IT should tell Blue Machines before making backend or data changes that affect the data used by the agent.

We should not find out about these changes only through an information-only ticket after the change has happened.

7. Add per-intent monitoring.

Add response-time monitoring for each intent, with alerts.

This problem took 3 weeks and an internal escalation to become visible.

But the data was available from the beginning.

The problem was that we were not looking at the response time by intent.